

1. Administrative & Study Identification

Full Study Title: A Phase III, Multicenter, Single-Arm Study Evaluating the Efficacy, Safety, Pharmacokinetics, and Pharmacodynamics of Crovalimab in Pediatric Patients With Atypical Hemolytic Uremic Syndrome (aHUS) **Short Title/Acronym:** COMMUTE-p **Protocol Number:** B042354 **NCT Number:** NCT04958265 **Posting ID:** 983203046551048668 **Sponsor/Company Name:** Roche **Investigational Product:** Crovalimab **Phase of Development:** Phase III **Trial Status:** ACTIVE_NOT_RECRUITING **Targeted Enrollment:** 41 **Disease Area:** Atypical Hemolytic Uremic Syndrome **Medical Monitor:** Roche Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To evaluate the efficacy of crovalimab in pediatric participants with aHUS by assessing the percentage of participants achieving a complete Thrombotic Microangiopathy (TMA) response (cTMAr) at Week 25 (after 24 weeks on treatment). * **Secondary Objectives:** To evaluate the safety, pharmacokinetics (PK), and pharmacodynamics (PD) of crovalimab, as well as kidney function stability, change in estimated glomerular filtration rate (eGFR), changes in dialysis status, changes in Chronic Kidney Disease (CKD) stage, and long-term hematologic parameters.

2.2 Background & Rationale To address the critical unmet medical need in pediatric patients with atypical Hemolytic Uremic Syndrome (aHUS), an ultra-rare, life-threatening condition caused by complement system dysregulation. Current treatments require frequent intravenous infusions. Crovalimab, a novel C5 inhibitor, offers a weight-based intravenous loading dose followed by convenient subcutaneous (SC) maintenance dosing, potentially reducing treatment burden while protecting red blood cells and preventing end-stage kidney failure.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Single-arm * **Blinding:** Open-label * **Randomization:** N/A (Single Group Assignment)

3.2 Planned Enrollment & Duration * **Target \$N\$:** 41 Pediatric participants (staggered by weight cohorts) * **Trial Status:** Active, Not Recruiting * **Number of Sites:** Multicenter (Three EU

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Pediatric patients ≤ 18 years old (infants must be at least 28 days old and weigh at least 5 kg at screening). 2. Confirmed diagnosis of atypical Hemolytic Uremic Syndrome (aHUS). 3. Vaccinated against *Neisseria meningitidis* serotypes A, C, W, Y, and B; *Haemophilus influenzae* type B; and *Streptococcus pneumoniae* according to national vaccination recommendations. 4. For female participants of childbearing potential: an agreement to remain abstinent or use contraception. 5. Participants with a prior kidney transplant are eligible if they have a known history of complement-mediated aHUS prior to the transplant. 6. For Naive Cohort: Onset of initial TMA presentation within 28 days prior to the first dose. 7. For Switch Cohort: Documented treatment with eculizumab or ravulizumab and clinical evidence of response. 8. For C5 SNP Pretreated Cohort: Poorly controlled TMA following treatment with another C5 inhibitor and known C5 polymorphism. 9. Stable dose of allowed concomitant therapies (e.g., immunosuppressants, corticosteroids, mTORi, or calcineurin inhibitors) for ≥ 28 days prior to screening.

4.2 Exclusion Criteria 1. Diagnosis of a condition leading to non-aHUS TMA (e.g., TTP, STEC-TMA, Pneumococcal HUS, TMA secondary to cobalamin C defect, or DGKE nephropathy). 2. Positive direct Coombs test. 3. Active systemic bacterial, viral, or fungal infection within 14 days before first crovalimab administration, history of *Neisseria meningitidis* infection within 6 months, known immune deficiency, or Positive HIV test. 4. Recent intravenous immunoglobulin (IVIg) treatment, chronic dialysis within 90 days prior to the first dose, end-stage renal disease, or multi-system organ failure. 5. Presence or history of a condition that could trigger TMA (e.g., malignancy, bone marrow or organ transplant other than kidney, autoimmune disease). 6. Normalization of serum creatinine values at baseline (≤ 97.5 th percentile for age) (Naive Cohort only). 7. Positive for active Hepatitis B and/or C infections (HBV/HCV) or cryoglobulinemia at screening (for Switch/Pretreated Cohorts recently receiving C5 inhibitors). 8. Pregnant or breastfeeding, or intending to become pregnant. 9. Recent use of tranexamic acid, or participation in another interventional trial within 28 days or five half-lives of an investigational product.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Investigational Product:** Crovalimab
* **Dosage/Frequency:** Initial weight-based Intravenous (IV) loading dose on Week 1 Day 1. This is followed by Subcutaneous (SC) doses

on Week 1 Day 2, and Weeks 2, 3, and 4. From Week 5 onward, maintenance weight-based SC doses are administered every 2 weeks (Q2W) or 4 weeks (Q4W) depending on participant weight. * **Route of Administration:** Intravenous (IV) for loading dose; Subcutaneous (SC) for maintenance doses. * **Duration of Treatment:** 24 weeks for the primary efficacy assessment, with continuous long-term treatment until study completion.

5.2 Concomitant & Prohibited Therapy * **Permitted:** Concomitant immunosuppressants, corticosteroids, mTOR inhibitors, or calcineurin inhibitors, provided the participant has been on a stable dose for ≥ 28 days prior to screening. * **Prohibited:** Current/previous treatment with other complement inhibitors (in the naive cohort), recent use of tranexamic acid, and participation in another interventional trial within 28 days. Specific exclusion guidelines exist for the timing of plasma exchange/infusions relative to crovalimab administration.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, aHUS Diagnosis, Vaccination verification, Baseline Labs
Baseline	Week 1 Day 1	First IV Loading Dose, PK/PD sampling, Safety check
Interim	Weeks 1 to 4	Subcutaneous (SC) induction doses (Day 2, Wk 2, Wk 3, Wk 4), Safety Labs
End of Primary Phase	Week 25	Primary Efficacy Assessment (Complete TMA response at 24 weeks)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Percentage of Participants with complete TMA response (cTMAr) in the Naive Cohort from Baseline up to Week 25 (after 24 weeks on treatment). *

Measurement Method: Laboratory analysis demonstrating normalization of platelet counts, lactate dehydrogenase (LDH), and $\geq 25\%$ improvement in serum creatinine from baseline.

7.2 Secondary & Exploratory Endpoints * Change from Baseline in Dialysis Status. * Change in Estimated Glomerular Filtration Rate (eGFR) (Naive and Switch Cohorts). * Percentage of Participants with Change from Baseline in Chronic Kidney Disease (CKD) stage (Naive and Switch Cohorts). * Percentage of participants with cTMAr in the Switch and Pretreated Cohorts. * Pharmacokinetics (PK) and Pharmacodynamics (PD) profiling of Crovalimab.

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Routine collection of AE data, focusing heavily on signs of systemic infection (particularly meningococcal infections) and injection-site reactions.
 - **Serious Adverse Events (SAEs):** Reported within 24 hours of site awareness, particularly concerning multi-organ failure or severe infections.
 - **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee to evaluate ongoing safety, given the pediatric population and ultra-rare disease indication.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** Intent-to-Treat (ITT) population for primary effectiveness outcomes; Safety Set for all participants receiving at least one dose of Crovalimab.
 - **Sample Size Justification:** Sized to detect meaningful clinical responses in an ultra-rare pediatric orphan disease cohort, with a targeted enrollment of 41 participants, staggered across different weight groups (e.g., <20 kg, $20-40$ kg, ≥ 40 kg).
 - **Handling of Missing Data:** Handled via prespecified imputation algorithms per the SAP; participants missing critical end-of-assessment labs without documented clinical justification are typically analyzed as non-responders for the cTMAr endpoint.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Scheduled onsite and remote monitoring visits to ensure protocol adherence, correct weight-based dosing, and strict monitoring of vaccination records prior to C5 inhibition.
 - **Audit Trail:** Robust electronic Case Report Form (eCRF) tracking, with detailed data clarification forms for any discrepancies in TMA diagnostic parameters.
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11. Study Identifiers & Governance

- **Primary ID:** B042354
- **Registry Number:** NCT04958265
- **Posting ID:** 983203046551048668
- **Sponsor/Lead Organization:** Roche
- **Study Title:** COMMUTE-p
- **Official Regulatory Title:** A Phase III, Multicenter, Single-Arm Study Evaluating the Efficacy, Safety, Pharmacokinetics,

12. Clinical Framework (aHUS Specific)

- **Primary Condition:** Atypical Hemolytic Uremic Syndrome (aHUS)
 - **Diagnosis Threshold:** Clinical and laboratory evidence of TMA (thrombocytopenia, hemolysis) and exclusion of STEC-HUS and TTP
 - **Medical Methodology:** Complement C5 Inhibition
 - **Optimization Target:** Complete TMA response (cTMAr)
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13. Study Procedures & Methodology

- **Management Pathway:** Standard C5 inhibitor pathway transitioning from IV loading to SC maintenance
 - **Education Protocol (HCP):** Training on proper weight-based dosing schedules and monitoring for signs of meningococcal infections.
 - **Education Protocol (Patient):** Caregiver training on at-home SC self-administration techniques and signs/symptoms of systemic infection requiring immediate medical attention.
 - **Quality Audit Mechanism:** Tracking of timely dose administrations (Q2W or Q4W compliance) and patient/caregiver adherence monitoring.
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14. Observation & Follow-Up Schedule

- **Total Duration:** Up to 8 years (Completion expected May 2029)
 - **Onsite Visit Cadence:** Intensive during Week 1 through 5, followed by scheduled visits aligning with Q2W or Q4W dosing assessments.
 - **Remote Monitoring Frequency:** Regular interval check-ins for safety reporting and SC administration compliance.
 - **Checklist Review Interval:** Every 4 weeks (Q4W) concurrent with maintenance dosing schedules.
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15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				